

Social Media Advertising Loads as Prices

NBER working paper

George Beknazar-Yuzbashev, Rafael Jiménez-Durán, Andrey Simonov, and Mateusz Stalinski

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Presenter: Haotong Zhang, SOE, RUC

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- 2 Theory Model
- 3 Empirical: field experiment
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Core question

- Why advertising dominates social media monetization?
- Specifically, why users "pay" with attention (ad-funded) rather than money (subscription-funded)?
- Three channels: existence? importance?

1. Attention-price sensitivity

Users may tolerate attention prices more than monetary prices.

2. Microtargeting

Better matching may make ads less costly or more useful to users.

3. Personalized ad loads

Individualized attention prices allow the platform to price discriminate.

- Framework: Theory Model → Field Experiment (estimation) → Counterfactual

Related literature

① Advertising as an attention price.

- theoretical model; negative relationship between ad loads and consumption; elasticity of time spent with respect to ad loads of 0.094.
- **This paper:** (1) varying both ad loads & match quality; (2) document personalized ad loads.(3 channels)

② Social media and platform monetization.

- ad-driven → harmful content; advertiser demand; regulatory scrutiny.
- **This paper:** Focus on the consumer side; rationalize the prevalence of ad-based business models on platforms.

③ Privacy.

- limiting tracking reduces advertising effectiveness and platform revenues; privacy paradox(concerns expressed, welfare cost limited).
- **This paper:** isolate the consumer response (worse targeting → user engagement ↓ but does $\Delta WTP \approx 0$).

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User side: choose t

Utility of becoming a membership of the platform:

$$v^\theta(a^\theta, q^\theta, x) = \max_{t \geq 0} u^{\theta,c}(t) + u^{\theta,a}(ta^\theta, q^\theta) + u^{\theta,j}(x) - w^\theta t(1 + a^\theta)$$
$$\Rightarrow t^\theta(a^\theta, q^\theta) = \operatorname{argmax}_{t \geq 0} v^\theta(a^\theta, q^\theta, x)$$

Total net utility: $\varepsilon + v^\theta(a^\theta, q^\theta, x) - s$; **Joining the platform** $\iff \varepsilon \geq s - v^\theta(a^\theta, q^\theta, x)$

- Type (θ, ε) : observable characteristics $\theta \sim G_\theta$ + membership benefits $\varepsilon \sim F_{\varepsilon|\theta}$;
- time spent on contents: t^θ , opportunity cost: w^θ (monetizing);
- **ad load:** $a^\theta \in [0, 1]$ (counted as time spent ratio per content);
 - $\implies$ Total time spent on ads: $A^\theta = a^\theta t^\theta$
 - $\implies$ Total time spent on platform: $A^\theta + t^\theta = (1 + a^\theta)t^\theta$
- **ad quality:** q^θ ;
- market share: $x \implies u^{\theta,j}(x)$ captures the network effect;
- subscription fee: s .

User side: demand

The demand for the platform of θ users is given by

$$X^\theta(a^\theta, q^\theta, s, x) = g^\theta \int_{s-v^\theta(a^\theta, q^\theta, x)}^{\infty} f_{\varepsilon|\theta}(\varepsilon \mid \theta) d\varepsilon$$

$$\implies \text{Total platform demand: } X(a, q, s, x) = \sum_{\theta} X^\theta(a^\theta, q^\theta, s, x)$$

Fixed point problem:

$$X(a, q, s, x) = x \implies \text{unique } x(a, q, s)$$

Platform: choose (a^θ, q^θ)

$$\pi(p, a, q, s) = \sum_{\theta} \left[\underbrace{p^\theta a^\theta t^\theta(a^\theta, q^\theta) x^\theta(a, q, s)}_{\text{ad revenue}} + \underbrace{s x^\theta(a, q, s)}_{\text{subscription revenue}} \right].$$

- $p^\theta(a^\theta, q^\theta)$: equilibrium ad price per unit of engagement with type- θ users (reflecting advertiser WTP).

Ad-based platform:

$$\begin{aligned} & \max_{a, q} \pi(p(a, q), a, q, s = 0) \\ (a^*, q^*) &= \operatorname{argmax}_{a, q} \pi(p(a, q), a, q, 0), \quad p^* = p(a^*, q^*). \end{aligned}$$

Subscription-based platform:

$$\begin{aligned} & \max_s \pi(p, a = 0, q, s) \\ s^* &= \operatorname{argmax}_s \pi(p, a = 0, q, s). \end{aligned}$$

Platform: relative profitability

$$\begin{aligned} & \pi(\bar{p}, a = 0, \bar{q}, s^*) - \pi(p^*, a^*, q^*, s = 0) \\ &= \underbrace{\pi(p^*, a^*, \bar{q}, s = 0) - \pi(p^*, a^*, q^*, s = 0)}_{\text{1. Reduced ad match quality}} \\ &+ \underbrace{\pi(\bar{p}, a^*, \bar{q}, s = 0) - \pi(p^*, a^*, \bar{q}, s = 0)}_{\text{2. No ad price differentiation}} \\ &+ \underbrace{\pi(\bar{p}, \bar{a}, \bar{q}, s = 0) - \pi(\bar{p}, a^*, \bar{q}, s = 0)}_{\text{3. No ad-load discrimination}} \\ &+ \underbrace{\pi(\bar{p}, a = 0, \bar{q}, s^*) - \pi(\bar{p}, \bar{a}, \bar{q}, s = 0)}_{\text{4. Business-model change}}. \end{aligned}$$

Components 1 and 2: both arise from removing microtargeting.

1 is the user-side match-quality effect; 2 is the advertiser-side loss of price differentiation.

Components 3 and 4: loss of personalized attention prices, then the direct switch to monetary pricing.

Equilibrium & Two key forces.

$$a^\theta = \frac{p^\theta t^\theta + \sum_{\tilde{\theta}} \frac{p^{\tilde{\theta}} a^{\tilde{\theta}} t^{\tilde{\theta}}}{x^\theta} \frac{\partial x^{\tilde{\theta}}}{\partial a^\theta}}{\left| \frac{\partial t^\theta}{\partial a^\theta} \right| p^\theta + \left| \frac{\partial p^\theta}{\partial a^\theta} \right| t^\theta}, \quad q^\theta = 1 (\text{corner solution})$$

- **Inverse elasticity:** Less ad-sensitive users $\Rightarrow$ higher optimal ad loads.

$$\left| \frac{\partial t^\theta}{\partial a^\theta} \right| \downarrow \Rightarrow a^\theta \uparrow$$

- **Network effect:** Larger positive spillovers $\Rightarrow$ lower optimal ad loads.

$$\frac{\partial x^{\tilde{\theta}}}{\partial a^\theta} < 0$$

- p^θ : *ambiguous*

Theory to Empirical

Core question 1: How does a & q affects users' behavioral consequences?

Theory: $t \implies$ Outcomes: y , including 4 aspects

- $\partial y / \partial a$: channel 1 (sensitivity to ad load)
- $\partial y / \partial q$: channel 2 (sensitivity to ad targeting)

Core question 2: Personalization of ad load? - channel 3

Theory: a^θ (θ ? inverse elasticity? network effect? p^θ ?)

$\implies$ demographic characteristics, predicted elasticity, platform friends, Ad value

Core question 3: How does the *ad-based* model differ from the *subscription-based* model?

Theory: $\Delta\pi \implies$ counterfactual t_i, v_i

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Recruitment and sample

- Recruited through **Facebook ads** from June - December 2024.
- Eligible users: U.S. adults, English-language Facebook users, primarily using Google Chrome.
- Enrollment required:
 - installation of the custom browser extension,
 - baseline and endline surveys (demographic characteristics + usage + WTA).
- Participants received a \$21 gift card upon completion.
- Final experimental sample: 1,810 users, with sufficient baseline activity and reliable extension tracking.

Experimental timeline

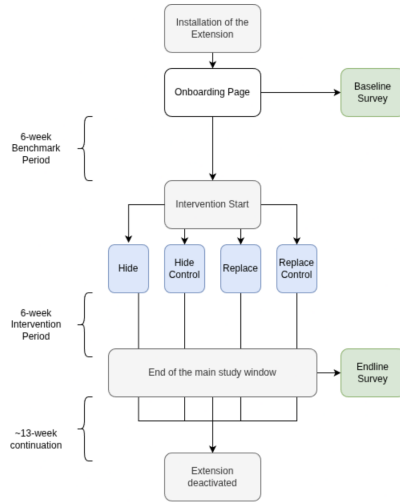


FIGURE A1: STUDY FLOW

Four experimental arms

Hide

Suppress the visual display of most ads.

Replace

Replace selected ads with ads shown to other users (one-for-one).

Hide Control

Leave the feed unchanged.

Replace Control

Run the same replacement procedure, but replace each ad with itself.

- Hide vs. Hide Control: isolates load.
- Replace vs. Replace Control: isolates targeting.

Measurements: ad load & targeting

① Ad load a

- **Experienced ad load:**

$$\frac{\text{\# ads observed}}{\text{\# ads} + \text{\# organic posts/comments observed}}$$

- **Supplied ad load:** same ratio, but *before* the intervention is applied.
- **Feed ad load:** ads as a share of content in the main news feed, excluding comment sections and other non-feed areas.

② Ad targeting / match quality q

- Construct an **ad distance** between user i and advertiser a :

$$d_{ia} = \sqrt{\sum_{k \in K} (X_{ik} - \bar{X}_{ak})^2},$$

where $K = \{\text{young, male, white, BA+}, \text{income} > 50K\}$.

- $\bar{X}_{ak}$: share of advertiser a 's baseline impressions delivered to users with characteristic k .
- Lower $d_{ia} \Rightarrow$ better ad-user match; higher $d_{ia} \Rightarrow$ worse targeting.
- Average d_{ia} across ads at the user-day level and normalize using the baseline mean and standard deviation.

Measurements: outcomes

Outcome family	Measures
Ad engagement	Impressions, clicks $\Rightarrow$ an ad-engagement index
Platform engagement	Active time, posts + comments consumed $\Rightarrow$ a platform-engagement index
Competing-platform use	X + Reddit, YouTube, other preregistered social media, and users' top-10 baseline domains
Platform valuation	WTA to deactivate Facebook for four weeks

- Active time captures interaction (allowing 3-mins brief pause).
- Day-level observation except WTA & demographic measures (only in baseline & endline surveys).

Data structure

DAILY PANEL

Six-week baseline
+ six-week intervention

Individual-day observations

Engagement, ad load, targeting, and
substitution

daily behavioral panel

identifies engagement effects

SURVEY PANEL

Baseline survey
+ endline survey at week 6

Two observations per person

WTA and demographic measures

two-period survey panel

identifies changes in WTA

Empirical strategy

Daily behavioral outcomes

$$Y_{it} = \alpha_i + \delta_{t,C_i} + \beta D_{it} + Errors_{it} + \varepsilon_{it}.$$

- α_i : individual fixed effects;
- δ_{t,C_i} : cohort $\times$ period fixed effects;
- D_{it} : treatment $\times$ intervention-period indicator.
- $Errors_{it}$: controls for browser-extension errors and their interactions with treatment.

Survey outcome: two-period DID

$$Y_{i,s} = \alpha_i + \lambda_s \text{Endline}_s + \delta(\text{Treat}_i \times \text{Endline}_s) + \varepsilon_{i,s}$$

Heterogeneous effects

$$Y_{it} = \alpha_i + \delta_{t,C_i,M_i} + \beta D_{it} + \phi(D_{it}M_i) + Errors_{it} + \varepsilon_{it},$$

where $M_i = 1$ for users with above-median baseline active time:

$$\beta = \text{effect for light users}, \quad \beta + \phi = \text{effect for heavy users}.$$

Identification assumptions and limitations

Experimental identification

- Random assignment after a common baseline
- Dedicated controls isolate:

$$\Delta a \quad \text{vs.} \quad \Delta q$$

- Strong first stages for ad load and targeting
- Direct behavioral measurement
- No meaningful treatment imbalance or differential attrition

Remaining scope / counterfactual assumptions

- Desktop, U.S., English-speaking recruited sample
- Heterogeneous Δa_i : instrumented by Hide assignment
- Mild endogenous platform response under Replace

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Channel 3: Is ad load heterogeneous across users?

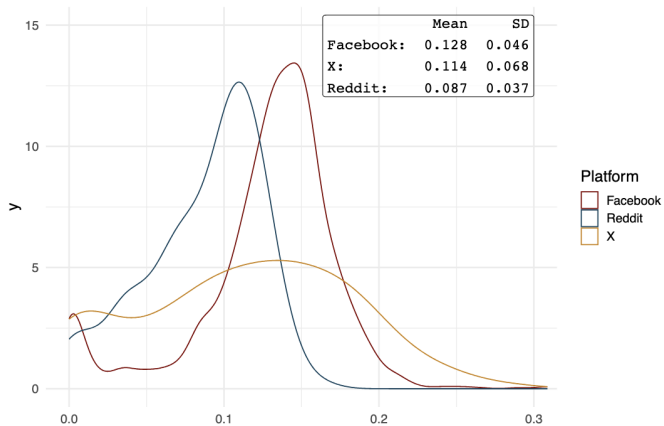
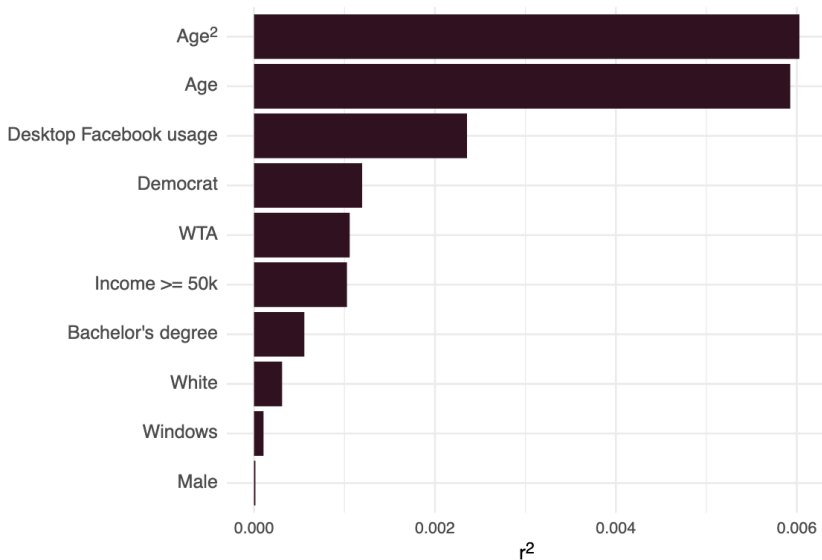
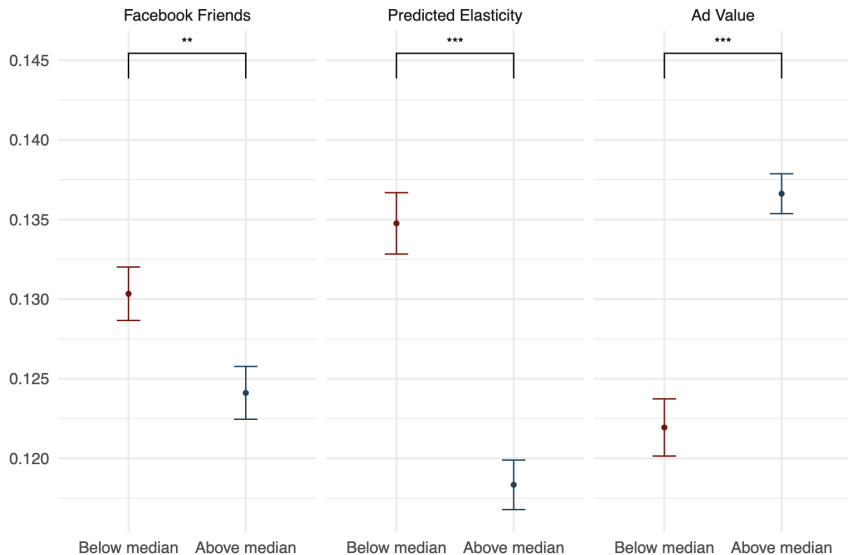


FIGURE 1: DISTRIBUTION OF INDIVIDUAL FEED AD LOAD BY PLATFORM

Channel 3: Is this heterogeneity systematic?



Channel 3: Is it consistent with the model's optimal pricing logic?



Channel 1: Users are highly insensitive to ad load

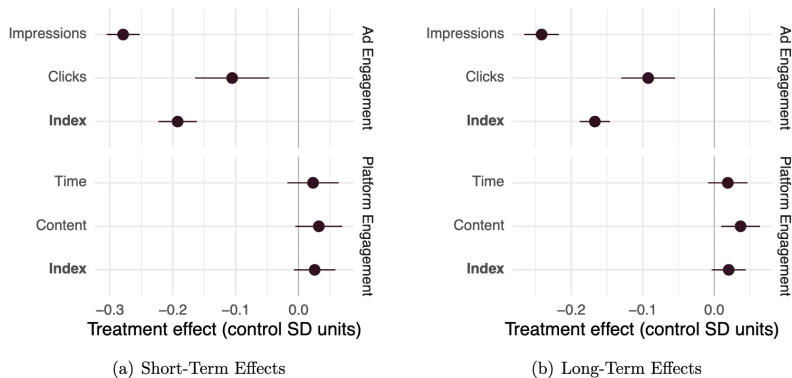


FIGURE 6: OWN-PLATFORM ENGAGEMENT EFFECTS OF HIDING INTERVENTION

-82% ad load $\Rightarrow$ $+6\%$ time, $+9\%$ content

Channel 2: Microtargeting improves both ad match and platform engagement

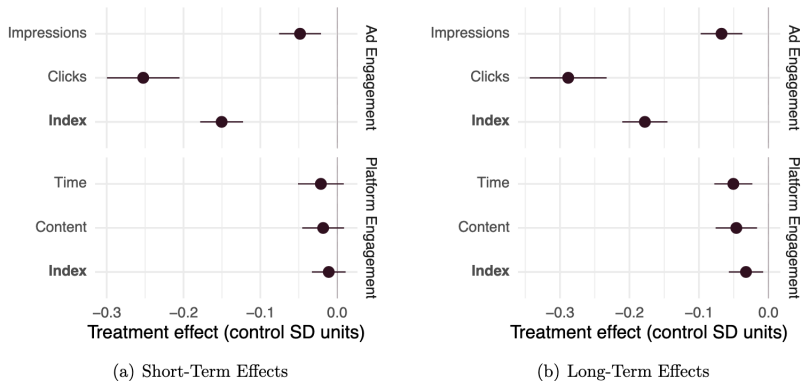


FIGURE 7: OWN-PLATFORM ENGAGEMENT EFFECTS OF REPLACEMENT INTERVENTION

$q \downarrow \Rightarrow \text{clicks} \downarrow \Rightarrow \text{long-run time/content} \approx -13\%$.

Channel 1 & 2 results: summary

Outcome	Hide	Replace
Ad engagement	-0.19 SD	-0.15 SD
Platform engagement: short run	Small $\uparrow$ Time +0.65 min Content +5.64	≈ 0 No significant change
Platform engagement: longer run	Small $\uparrow$, persistent Content +6.63/day	$\downarrow\downarrow$ Time/content $\approx -13\%$
Competitor use	Little substitution	No clear substitution
WTA / platform valuation	≈ 0	≈ 0
Who responds most?	Heavy users: larger ad-side responses	Heavy users: strongest long-run engagement losses

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What is the counterfactual asking?

A. Status quo

Personalized ad loads and
microtargeted ads

$$a^* \rightarrow \bar{a}, q^* \rightarrow \bar{q}$$

B. Pure subscription

Uniform fee and no ads

$$\pi(p, a = 0, q, s)?$$

C. Hybrid opt-in

Ad-supported version or
paid ad-free tier

Which regime best balances profits, engagement, demand, and consumer surplus?

The four-way decomposition of $\Delta\pi$ in Eq. (1) is mainly the transition from A to B.
The opt-in regime is an additional policy counterfactual.

Procedures

- **Step 1: Estimate heterogeneous causal responses**

- Collapse the daily panel to user-level intervention-minus-baseline changes.
- Estimate user-specific responses to ad load and targeting:

$$\hat{\gamma}_i^a, \quad \hat{\gamma}_i^q.$$

- Use the Hide assignment as an instrument for realized changes in ad load, and the Replace experiment to estimate targeting effects.

- **Step 2: Construct counterfactual behavior**

- Use a local-linear approximation:

$$y_i(a'_i, q'_i) = y_i + \hat{\gamma}_i^a(a'_i - a_i) + \hat{\gamma}_i^q(q'_i - q_i).$$

- Obtain counterfactual platform engagement and valuation, including

$$t_i(a'_i, q'_i), \quad WTA_i(a'_i, q'_i).$$

Procedure

- **Step 3: Add valuation and network effects**

- Calibrate the effect of aggregate platform participation on user valuation.
- Convert willingness to accept into willingness to pay:

$$WTP_i = 0.05 WTA_i.$$

- For each candidate subscription price s , solve the participation fixed point:

$$x = X(a', q', s, x).$$

- **Step 4: Solve and compare monetization regimes**

- Compute equilibrium outcomes for each candidate s :

$$x(s), \quad T(s), \quad CS(s), \quad \pi(s).$$

- Choose the profit-maximizing subscription fee:

$$s^* = \arg \max_s \pi(s).$$

- Compare the status-quo personalized advertising model with the optimal subscription regime, and decompose the difference using Equation (1).

$$\text{RCT} \rightarrow (\hat{\gamma}_{i,y}^a, \hat{\gamma}_{i,y}^q)$$

$$\Downarrow$$

$$y_i(a', q') = y_i + \hat{\gamma}_{i,y}^a(a'_i - a_i) + \hat{\gamma}_{i,y}^q(q'_i - q_i)$$

$$\Downarrow$$

$$y \rightarrow \begin{cases} t_i(a', q') \\ WTA_i(a', q') \end{cases} \rightarrow v_i(a', q', x)$$

$$\Downarrow$$

$$x = X(a', q', s, x) \rightarrow x(s)$$

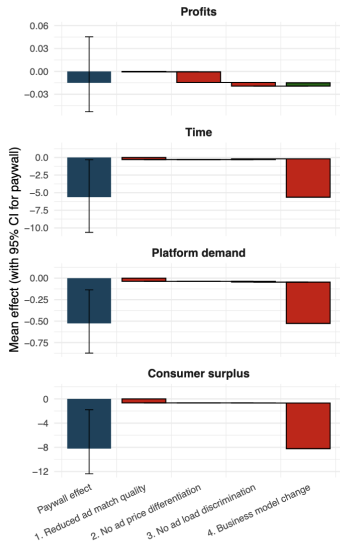
$$\Downarrow$$

$$\{t_i, v_i, x_i\}_i \xrightarrow{\text{aggregation}} \{\text{Time}, CS, x, \pi\}, s^* = \arg \max_s \pi(s)$$

$$\Downarrow$$

$$\Delta Y = Y_{\text{Subscription}} - Y_{\text{Ads}} = C_1 + C_2 + C_3 + C_4$$

Results: Regime 1 & 2



Recall:

$$\begin{aligned}
 & \pi(\bar{p}, a = 0, \bar{q}, s^*) - \pi(p^*, a^*, q^*, s = 0) \\
 &= \underbrace{\pi(p^*, a^*, \bar{q}, s = 0) - \pi(p^*, a^*, q^*, s = 0)}_{\text{1. Reduced ad match quality}} \\
 &+ \underbrace{\pi(\bar{p}, a^*, \bar{q}, s = 0) - \pi(p^*, a^*, \bar{q}, s = 0)}_{\text{2. No ad price differentiation}} \\
 &+ \underbrace{\pi(\bar{p}, \bar{a}, \bar{q}, s = 0) - \pi(\bar{p}, a^*, \bar{q}, s = 0)}_{\text{3. No ad-load discrimination}} \\
 &+ \underbrace{\pi(\bar{p}, a = 0, \bar{q}, s^*) - \pi(\bar{p}, \bar{a}, \bar{q}, s = 0)}_{\text{4. Business-model change}}.
 \end{aligned}$$

- Which mechanisms actually explain the gap between ads and subscription?
- Profits: component 2, advertiser-side price differentiation effect.
- Other outcomes: component 4, business model change.

Further results

- **Why does the business-model change matter so much?**

Appendix Figure A12 decomposes Component 4 into:

- relative revenue per user,
 - sensitivity to monetary prices,
 - sensitivity to ad loads.
- Subscriptions generate **higher revenue per participating user**, but users are much **more sensitive to monetary prices** than to advertising intensity.
- **Trade-off:** *declining participation and engagement largely v.s. per-user revenue advantage of subscriptions;*
- Network effects further amplify the participation loss.
- Component 1, the user-side value of microtargeting, and Component 3, eliminating ad-load discrimination by setting $a_i = \bar{a}$, are not the main drivers of the gap between the advertising and subscription business models.

3 Channels	Section 4 evidence	Figure 12 mapping and quantitative role
1. Low sensitivity to ad load	User side: Hide sharply reduces a , but produces only small increases in t and content.	Mainly Component 4: business-model change. Users are much more sensitive to monetary prices than to attention prices, driving the differences in engagement, demand, and consumer surplus.
2. Microtargeting	User side: Replace worsens match quality, reducing clicks and long-run platform usage.	Component 1: user-side match quality — quantitatively modest. Component 2: advertiser-side ad-price differentiation — the main source of the profit advantage of advertising.
3. Personalized ad loads	Persistent cross-user heterogeneity in a_i ; ad-load allocation is consistent with inverse-elasticity and network-spillover predictions.	Component 3: no ad-load discrimination , $a_i \rightarrow \bar{a}$. Clearly present empirically, but quantitatively smaller in the overall ads-versus-subscription gap.

Regime 3

- **Hybrid opt-in model:** users may choose between the ad-supported platform and a paid ad-free subscription.
- Only about **17%** of users choose the subscription option in the simulation, so the hybrid regime produces only negligible changes in profits, engagement, platform demand, and consumer surplus relative to the status quo.

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Conclusion

Main takeaway

- Advertising remains competitive with subscription because users are **much less sensitive to attention prices than to monetary prices**, while microtargeting additionally improves monetization through **advertiser-side price differentiation**.

What I learned from this paper

- **Experiment guided by theory:** start from the model, identify the key behavioral objects (a, q, t, v) , and design separate interventions to isolate ad-load and targeting effects.
- **Identification before counterfactual:** use randomized variation to estimate heterogeneous causal responses, then map these estimates into economically meaningful counterfactual objects.
- **A flexible counterfactual strategy:** combine local-linear extrapolation, external calibration, and an equilibrium participation fixed point rather than estimating a fully structural model.
- **Theory disciplines empirical design:** the model clarifies which margins must be identified, what variation is required, and which additional assumptions are needed for extrapolation.

Main limitation / extension

- The paper is not a conventional fully structural two-sided-market estimation: the advertiser side is relatively parsimonious, and strategic interactions between users, advertisers, and the platform are largely outside the estimated system.
- A natural extension is to combine the paper's experimentally identified consumer-side responses with a richer advertiser-demand and equilibrium ad-pricing framework, potentially building on Katz and Allcott (2025).